MILITARY SPECIFICATION

SCRAPER, ELEVATING, SELF-PROPELLED,

9 CUBIC YARD MINIMUM HEAPED-CAPACITY,

SECTIONALIZED AND NON-SECTIONALIZED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air transportable, air-droppable, helicopter transportable, diesel-engine-driven, elevating scrapers.

1.2 Classification. The scrapers shall be of the following types, as specified (see 6.2):

Type I - Non-sectionalized (Air Transportable and Air-droppable).
Type II - Sectionalized (Air Transportable, Air-droppable, and Helicopter Transportable).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	Lubricating Oil, Internal Combustion.
MIL-L-2105	Lubricating Oil, Gear, Multipurpose.
MIL-T-3351	Tractor, Full-Tracked, Low Speed; Tractors Wheeled, Agricultural; and Tractor, Wheeled, Industrial; and Their Attachments, Packaging of.

FSC 3805

Beneficial cements (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-7866	Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52798	Enamel, Alkyd, Camouflage.
MIL-D-52987	Distributor, Water, Self-Propelled, Minimum 2500 Gallon, Sectionalized and Non-Sectionalized.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-669	Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel.
MIL-STD-814	Requirements for Tiedown, Suspension and Extraction Provisions on Military Materiel for Airdrop.
MIL-STD-1188	Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	Methods for Selection of Industrial Engines for End Item Application.

MILITARY HANDBOOK

AIR FORCE SYSTEM COMMAND
Design Handbook DHI-11 - Air Transportability.

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HK, Fort Belvoir, VA 22060.)

3. REQUIREMENTS

3.1 General. The scraper shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public for the same or similar application, interfacing with the same or similar components, except as specified herein, for at least 1 year prior to invitation for bids or request for proposal. Product improvements incorporated in the commercial production units in this 1 year period are acceptable. In the case of a recently introduced item, the contractor shall submit sufficient valid industrial operation/test data including all test reports for the item and its components, such as engine, transmission and hydraulic systems field user reports, including any failures or problem areas, and corrective action taken to demonstrate acceptability of the item. When specified requirements are not stated herein, all items listed as standard equipment in the contractors published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished, The scraper shall be equipped with any optional equipment necessary to comply with requirements specified herein, Optional equipment is defined as equipment not standard with the scraper but which has been furnished to the commercial customer, such as special features or allied equipment. The scraper shall be equipped with all components necessary to enable the scraper to function reliably and efficiently in sustained operation. The scraper shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect on the date of invitation for bids or request for proposal. The tractor section shall be interchangeable with the tractor section of the water distributor conforming to MIL-D-52987. The interchangeability shall be accomplished at the user level without cutting or welding.

3.2 Description. The scraper shall be a commercial elevating model, with diesel engine, pneumatic tires, two axles, articulated steering, two single driving front wheels, and two single non-driving gearwheels. The scraper shall have a minimum heaped capacity of 9 cubic yards, and shall be equipped with power shift transmission.

3.2.1 Type I. The type I scraper with ROPS/FOPS removed shall be capable of being loaded and rigged on an air delivery platform in accordance with MIL-STD-669, air transported and air delivered by parachutes.

3.2.2 Type II. The type II scraper with ROPS/FOPS removed shall meet the requirements specified for type I (see 3.2.1) and shall also be capable of being sectionalized into 2 sections for external air transport by helicopter Sectionalization material may be removed for airdrop

3.3 First article (preproduction models). The contractor shall furnish two scrapers of each type for examination and demonstration within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce scraped that comply with the requirements of this specification. The examination and demonstrations shall be

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as specified in Section 4. Following successful completion of the examination and demonstration by the contractor, the Government shall perform a simulated airdrop demonstration on one of each type scraper at a site selected by the Government. The examination and all demonstrations conducted by the contractor shall be subject to surveillance and approval by the Government. Acceptance of the preproduction models will be-dependent on successful completion of the examination, contractor demonstrations, and the Government conducted simulated airdrop demonstration.

3.4 Production scrapers. All scrapers of the same type furnished under this specification shall be physically and mechanically identical to the first article scraper approved by the Government.

3.5 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.6 Safety. All available standard or optional accessories shall be included to provide protection for maintenance and operating personnel from rotating or reciprocating parts and from all parts subject to high operational temperatures and so located as to be a hazard. Exhaust discharges shall be directed so as to not endanger personnel. All olatforms and steps shall have anti-skid surfaces. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the scraper can be set into motion. Contractor's standard grab irons shall be provided. A decal shall be furnished and shall be located in the area of the articulation point, warning of limited personnel clearence during frame articulation and placement of safety pin or bar for the scraper operating and transport modes.

3.6.1 Sound level. The exterior sound level of the scraper shall not exceed 88 dB(A) when measured in accordance with SAE J88. If the sound level in the area occupied by the operator exceeds 85 dB(A) when measured in accordance with SAE J919, a warning plate containing the following shall be permanently affixed to the scraper in a conspicuous protected location.

"WARNING, EAR PROTECTION REQUIRED WHEN OPERATING THIS SCRAPER"

The plate shall have a yellow background with black lettering and shall be on corrosion-resistance material.

3.6.2 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The scraper shall be equipped with a removable bolted down ROPS/FOPS canopy. The canopy shall be removable and replaceble without cutting or welding. The ROPS/FOPS shall conform to SAE J1040 and SAE J231 respectively. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. Quick disconnect fittings (electric, etc.) shall be provided as applicable to facilitate canopy removal. A sling eye(s) shall be provided to facilitate removal of ROPS/FOPS.

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3.7 Physical characteristics.

3.7.1 Weight limit. The basic weight of the scrapers, type I and type II, shall not exceed 30,600 pounds. Basic weight shall be determined with the ROPS/FOPS removed without any special items required for sectionalization, but with all other accessories, and with coolant, lubricating and hydraulic systems filled to capacity, and with not less than 25 percent nor more than 75 percent fuel. The type II scraper when sectionalized into 2 sections shall weigh not more than 16,000 pounds per section.

3.7.2 Dimensions. The overall width of the scraper shall not exceed 98 inches, except that the width may be increased to not more than 105 inches to accommodate special airborne/airmobile provisions. The overall length shall not exceed 396 inches. In order to conform to a height limit of 100 inches when rigged for airdrop, the reducible height of the scraper should be not more than 91 inches. No major component removal except ROPS/FOPS canopy, shall be required to obtain the reduced height. If the height exceeds 91 inches, the contractor must demonstrate prior to step I proposal opening date that his scraper can be rigged so as to conform to the 100 inch limit, and withstand the forces encountered in the simulated airdrop impact test (see 4.5.2.11) without damage or permanent deformation.

3.8 Sectionalization. The type II scraper less ROPS/FOPS, shall be capable of being sectionalized into two sections for helicopter transport. The weight of either section shall not exceed 16,000 pounds. The powered section shall be driveable and steerable under its own power for a distance of 50 feet over relatively flat unimproved areas. Reassembly time with tools furnished with the

shall not exceed 4 hours. Cutting and welding are not permitted. Components and accessories used in sectionalization and reassembly shall be stored on the scraper or in a remountable container attached to the scraper but shall not interfere with the operation of the scraper.

3.9 Performance. The scraper shall withstand the usage encountered in military operations such as: self-loading, hauling, dumping and spreading soil, crossing rough terrain, fording streams, and negotiating steep longitudinal and sidehill slopes, without damage. The scraper shall be capable of driving at maximum travel speed in transport, being towed by another vehicle, and limited towing of other equipment. The scraper shall be capable of self-loading undistributed or recompacted soil to heaped capacity in less than one minute.

3.9.1 Environmental conditions. Without the use of any external means, equipped with only standard/optional equipment, the engine shall start within five (5) minutes and the scraper shall-operate under full-load conditions within fifteen (15) minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be intergal with the scraper and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The scraper shall be capable of operating in hot humid tropical environments and dry dusty environments.

3.9.2 Travel speed. The scraper with heaped load shall be capable of a forward travel speed of not less than 20 mph on level pavement.

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3.9.3 Stability. The scraper shall negotiate 35 percent side slopes empty, 20 percent side slopes loaded, and descend 40 percent longitudinal slopes loaded without any wheel of the scraper losing contact with the ground. All parts requiring lubrication shall be lubricated when the scraper is operating under these conditions.

3.9.4 Fording. The scraper shall start and operate during and after fording operations in fresh water to any depth up to 20 inches including wave action without damage and without leakage of water into reservoirs containing lubricants.

3.10 Engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The engine shall be the engine furnished commercially including all systems, components, accessories, and auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow.

3.10.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816 shall be not less than 150 horsepower.

3.10.2 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.9.1 shall be provided.

3.10.3 Fuel tank. The fuel tank shall be of a type regularly furnished as standard equipment with a capacity for 10 hours of continuous normal operation. The tank shall be equipped with a means for draining water and sediment from the bottom of the tank.

3.10.4 Air compressor. An engine driven air compressor shall be furnished to provide air for the brake system (when equipped with air brake system) and for water distributor nozzle actuation. In addition to the air compressor, an air reservoir with safety valve, and the lines and fittings required for water distributor operation shall be provided on all tractors to accommodate the inter-changeability requirement of paragraph 3.1. A service outlet with hose and fittings necessary for tire inflation shall be provided with the air system.

3.11 Drive train. A drive train consisting of a torque converter, full powershift transmission, front drive axle with differential and planetary final drives, and all necessary connecting components shall be furnished. The transmission shall provide not less than four forward speeds and one reverse speed. A locking or limited-slip differential shall be furnished in the drive axle.

3,12 Wheels and tires. Commercial standard remountable wheels or rims shall be provided. Traction type tubeless tires shall be furnished and shall have a tread pattern suitable for scraper operation. The contractor's standard tire size and ply rating shall be furnished.

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3.13 Steering and braking,

3.13.1 Steering system. The scraper shall have an articulated frame type steering system which shall provide full control of the scraper under all normal operating conditions. The scraper shall be equipped with an emergency steering system conforming to SAE J53 that will give the operator steering control in the event of engine failure.

.3.13.2 Braking system. The scraper shall be equipped with power assisted brakes on all four wheels. A parking brake and an emergency braking system shall be provided. The braking system shall conform to SAE J1152.

3.14 Operator's compartment.

3.14.1 Seat. The contractor's standard commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone and all instruments and gages mounted in the operator’s compartment when seated with seat belt fastened and properly adjusted.

3.14.2 Instruments and indicators. The following instructments and indicators shall be provided:

(a) Engine oil pressure.
(b) Engine coolant temperature.
(c) Transmission/torque coverter oil temperature (if offered).
(d) Ammeter or voltmeter.
(e) Engine hourmeter or service meter.
(f) Air pressure gage.
(g) Low air pressure warning buzzer or horn (when equipped with air brake system).
(h) Fuel level indicator (gage or dipstick).
(i) Hydraulic oil level indicator (dipstick, sight tube or gage) (if offered).
(j) Air cleaner restriction indicator.
(k) Odometer (if offered).
(1) Speedometer or tachometer (if offered).

Additional instruments and indicators standard with the contractor’s commercial practice, which contribute to the safety and reliability of scraper operation, shall be provided.

3.14.3 Controls. All controls for the scraper shall be located in the operator's compartment and shall be within reach of the operator in his normal operating position. Mounting provisions for the water distributor air controls shall be provided on all tractors to accommodate the interchangeability requirements of 3.1.

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3.15 Hydraulic system. The scraper shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all scraper components

3.16 Electrical system. The contractor's standard 24-volt electrical system shall be provided.

3.16.1 Battery. The battery or batteries shall be the contractor~s standard or optional batteries for cold weather starting as specified in 3.9.1.

3.16.2 Alternator. The contractor's high output option alternator shall be furnished. The rated output shall be not less than 35 amps.

3.16.3 Lighting. The following lights with associated circuits, circuit breakers, or other protective devices, and switches shall be provided, if available as standard or as an option:

(a) Two headlights.
(b) Two front working lights.
(c) Combination stop and taillights.
(d) Turn signals with emergency flasher.
(e) Rear floodlight(s).
(f) Instrument panel lighting.

3.16.4 Electromagnetic radiation. The scraper shall comply with the electro-magnetic radiation limits of SAE J551.

3.16.5 Horn. A conventional operator controlled horn shall be furnished.

3.16.6 Back-up alarm. Aback-up alarm conforming to SAE J994 shall be furnished.

3.17 lowing devices. Tow hooks or other suitable means of attaching a tow cable or tow bar shall be provided on the front and rear of the scraper.

3.18 Bowl." The bowl shall be capable of power down, and shall provide for a minimum depth of cut of 6 inches. The cutting edge shall be either fixed or integral with the retractable floor, and equipped with replaceable, reversible wear plates. A minimum of four cutting edge teeth shall be provided. The bowl shall be equipped with reversible and interchangeable side bits.

3.18.1 Capacity. The bowl shall have a minimum heaped capacity of 9 cubic yards when measured in accordance with SAE J957.

3.18.2 Sliding floor/ejector. The combination sliding floor/ejector shall be equipped with double acting cylinders and shall provide for complete dumping of the bowl.

3.18.3 Elevator. The elevator shall behydrauli.tally driven, and shall provide at least two speeds forward and one speed reverse. A minimum of 15

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flights shall be provided. The elevating mechanism shall provide for ease of raising and lowering to aid in aircraft loading operations. Instructions for raising, lowering, and maintaining the elevator shall be provided in the technical publications.

3.19 Transportability. The scraper shall be capable of being transported by surface, rail, ocean-going carriers, and C130, C141, and C5A aircraft. The scraper shall withstand the impact forces encountered in air and rail shipments without damage or permanent deformation. The scraper shall conform to MIL-STD-669 for airdrop from C-130 aircraft and shall withstand stresses imposed when tested in accordance with 4.5.2.11 without breakage or permanent deformation of any structural member or component, or loss of ability of the scraper to perform all of its intended functions. When rigged for airdrop in accordance with MIL-STD-669, the scraper shall meet the established tipoff curves for C-130 and C-141 aircraft as defined in AFSC Design Handbook DHI-11. The scraper shall be equipped with slinging provisions and shall be capable of being lifted without damage using slings. The scraper shall be equipped with tiedown provisions for transport by the above modes. Suspension and extraction parachute provisions shall be provided for parachute delivery. Complete diagrams and instructions for lifting the scraper and tying down the scraper on freight cars and truck trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport. The contractor shall provide sufficient data to the Government with the step I proposal to confirm that the scraper being supplied can be loaded and transported in the C130, C141 and C5A aircraft. A steering frame lock in accordance with SAE J276 shall be provided. Removal of ROPS and other components to reduce overall height or width shall not prevent the scraper

3.19.1 Parachute suspension provisions. Four intergral suspension provisions conforming to MIL-STD-814 shall be furnished on the scraper. The parachute suspension provisions shall withstand the stresses in the amount and direction of pull specified without weld failure or permanent deformation of the scraper or the provision.

3.19.2 Slinging provisions. The scraper shall be provided with four slinging Provisions conforming to MIL-STD-209. In addition, each section of the type II scraper shall be provided with four slinging provisions conforming to MIL-STD-209. The provisions shall enable the scraper to be lifted in the normal traveling or operating position. Provisions shall-be positioned so that the lifting strain shall be in line with the longitudinal axis of the eye of the provision. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the scraper or the provisions. Parachute suspension provisions may be used as slinging provisions if they meet the requirements of MIL-STD-209.

3.19.3 Tiedown provisions. The scraper shall be provided with integral Provisions to permit tiedown of the scraper to the floor Or deck Of the tranPOr-tation medium or the airdrop platform. The tiedown provisions shall conform to MIL-STD-209 and MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the scraper or the provision.

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3.19.4 Exaction parachute provision. A single fitting shall be provided on the scraper for attachment of the parachute extraction system. Except as specified herein, the fittings shall conform to MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the scraper or the provisions. In lieu of the limit load specified in MIL-STD-814, the limit load of the provision shall be 56,580 pounds. The design ultimate strength of welded sections shall be four times the limit load.

3.20 Identification plates and marking. Each scraper shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, serial number, weight, date of manufacture and date of delivery. The identification plate shall be permanently affixed to the scraper in a readily accessible and conspicuous location. plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. All components removed or disassembled for shipment shall be matchmarked for proper reassembly. The scraper shall be-identified in accordance with the contractor's standard method.

3.21 Instruction plates. The scraper shall be equipped with instruction plates including warnings and cautions suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the scraper. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. Instructions for disassembly of the type II scraper into two sections and reassembly shall be provided on cards approximately 8-1/2 inches by 11 inches which shall be encased in laminated plastic. One set shall be furnished with each type II scraper

3.22 Shipping data plates. Shipping data plates for each scraper shall be furnished and shall show the silhouette of the scraper and each section (type 11 scraper only), indicating the location and capacity of tiedown provisions, slinging provisions, extraction provisions, suspension provisions, the center of balance and shipping weight. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. The plates shall be attached to the left side of the scraper and shall show the locations as viewed from the left side. The plates on the sections, (type II scraper only) shall be located in conspicuous protected locations. All plates shall be permanently attached.

3.23 Treatment and painting. The portions of the scraper normally painted shall be cleaned, treated, primed and painted in accordance with the contractor's standard commercial practice. The finishing coat shall be in accordance with MIL-E-52798, forest green color.

3.24 Stenciling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown, slinging, suspension, and extraction provisions, tire pressure and the center of gravity for type I and type II scraper, and the sectionalization storage area on the type II scraper.

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3.25 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each scraper shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924, and MIL-G-23827 as applicable.

3.26 Technical publications. Such technical publications as are specified (see 6.2), shall be furnished.

3.27 Special tools. Any nonstandard or special tools required for equipment operation and operator maintenance of the scraper shall be furnished with the scraper.

3.28 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.29 Technical information package. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2), depicting the scraper and its components exactly as it will be furnished. The Technical Information Package shall be submitted with all bids and proposals.

3.30 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.3):

(a) Container NSN 7520-00-559-9618 complete with maintenance forms. (The contractor shall furnish the fasteners and install the containers on the scraper.)
(b) Supplemental operating, maintenance and repair parts instructions.

3.31 Basic issue item. When specified (see 6.2), each item in the Basic Issue Item List shall be furnished with each scraper.

4. QUALITY ASSUREANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 Preproduction inspection. The first article (preproduction model) scrapers (see 6.2), furnished by the contractor shall be examined (see 4.3.1), and demonstrated (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and demonstration of the scrapers, the following shall be performed:

(a) Service the scrapers with the lubricants specified.
(b) Break in the scrapers as prescribed by the contractor.

4.3.1 Preproduction examination. Prior to the preproduction demonstration, the scrapers shall be examined for the defects marked "X" in the column 1 of table I. Presence of one or more defects shall be cause for rejection. Prior to the preproduction demonstration, the scrapers shall be examined and compared with this specification, with the contractor's published brochures, with commercial manuals, and with the completed Technical Information Package. Any discrepancy between the specification, brochures, manuals, or Technical Information Package and the preproduction models shall be cause for rejection. Prior to the preproduction demonstration, the certifications, and certified test reports required by 4.5.2.12 and 4.5.2.13 shall be examined. Incomplete, inaccurate, or otherwise unacceptable certifications, or certified test reports shall be cause for rejection.

4.3.2 Preproduction demonstration. Upon successful completion of the examination specified in 4.3.1, the preproduction model scraper shall be subject to the demonstrations marked "X" in column 1 of table II. Failure of any demonstration or any demonstration reports shall be cause for rejection.

4.4 Quality conformance inspection,

4.4.1 Individual examination. After successful completion of the demonstration specified in 4.4.2, each scraper shall be examined for the defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual demonstation. Each scraper shall be demonstrated as specified in column 2 of table II. Failure of any demonstration shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table I.

4.5.2 Demonstrations. Demonstrations shall be conducted in accordance with table II in the sequence listed.

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TABLE I. Examination schedule.

Preproduction Model(s)	Individual	Defect		Requirement Paragraph
1	2	3		4
X	X	101.	Scraper type not as specified.	1.2
X	X	102.	Scraper description not as specified.	3.2
X	X	103.	Safety characteristics not as specified.	3.6
X	X	104.	Articulation warning decal not furnished.	3.6
X	X	105.	Sound level warning plate not as specified.	3.6.1
X	X	106.	ROPS/FOPS not furnished or equipped with disconnects and lift eye.	3.6.2
X	-	107.	Dimensions not as specified.	3.7.2
X	X	108.	Sectionalization and components not as specified.	3.8
X	X	109.	Engine not as specified.	3.10
X	X	110.	Fuel tank not as specified.	3.10.3
X	X	111.	Drive train not as specified.	3.11
X	X	112.	Wheels and tires not as specified.	3.12
X	X	113.	Steering not as specified.	3.13.1
X	X	114.	Braking system not as specified.	3.13.2
X	X	115.	Seat not as specified.	3.14.1
X	X	116.	Instruments not as specified.	3.14.2
X	X	117.	Controls not as specified.	3.14.3
X	X	118.	Hydraulic system not as specified.	3.15
X	X	119.	Electrical system not as specified.	3.16
X	X	120.	Batteries not as specified.	3.16.1
X	X	121.	Alternator not as specified.	3.16.2
X	X	122.	Lighting not as specified.	3.16.3
X	X	123.	Horn not as specified.	3.16.5
X	X	124.	Back-up alarm not as specified.	3.16.6
X	X	125.	Towing devices not as specified.	3.17
X	X	126.	Bowl not as specified.	3.18
X	X	127.	Sliding floor ejector not as specified.	3.18.2
X	X	128.	Elevator not as specified.	3.18.3
X	-	129.	Transportability not as specified.	3.19
X	X	130.	Identification plates not as specified.	3.20
X	X	131.	Instruction plates not as specified.	3.21

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TABLE I. Examination schedule. (Cont'd)

Preproduction Model(s)	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	132. Shipping data plates not as specified.	3.22
X	X	133. Treatment and painting not as specified.	3.23
X	X	134. Stenciling instructions not furnished.	3.24
X	X	135. Lubrication not as specified.	3.25
X	X	136. Technical publications not as specified.	3.26
X	X	137. Special tools not furnished.	3.27
X	X	138. Toolbox not furnished.	3.28
X	-	139. Technical information package not as specified.	3.29
X	X	140. Government furnished containers not installed on scraper.	3.30
X	X	141. Basic issue items (if specified) not furnished.	3.31
X	X	142. Parts or components missing.	Section 3
X	X	143. Assembly incomplete or incorrect.	Section 3
X	X	144. Cracked or broken parts or assemblies.	Section 3
X	X	145. Permanent deformation of any part or assembly.	Section 3
X	X	146. Leaks in coolant, fuel, oil lines, or cylinders.	Section 3
-	X	147. Material not of same quality used on preproduction scrapers.	Section 3
X	X	148. Scraper not as specified.	Section 3
X	X	149. Compressor not as specified.	3.10.4

TABLE II. Demonstration schedule.

Preproduction Models		Individual	Test	Test Paragraph	Requirement Paragraph
1		2	3	4	5
Type I	Type II				
X	X	X	Production run-in	4.5.2.1	Section 3
-	X	-	Sound level	4.5.2.2	3.6.1
X	X	-	Weight	4.5.2.3	3.7.1 and 3.8
X	X	-	ROPS removal	4.5.2.4	3.6.2 and 3.19

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TABLE II. Demonstration schedule.

Preproduction Models		Individual	Test	Test Paragraph	Requirement Paragraph
1		2	3	4	5
Type I	Type II				
-	X	-	Sectionalization and interchangeability	4.5.2.5	3.1, 3.8 and 3.18.3
X	X	-	Parachute suspension provisions	4.5.2.6	3.19.1
X	X	-	Slinging provisions	4.5.2.7	3.19.2
X	X	-	Tiedown provisions	4.5.2.8	3.19.3
X	X	-	Extraction provisions	4.5.2.9	3.19.4
-	X	-	Construction task	4.5.2.10	3.9 and 3.10.3
X	X	-	Airdrop	4.5.2.11	3.19

4.5.2.1 Production run-in. Start and operate each scraper as follows:

Operate the scraper in all gears forward and reverse at maximum speeds commensurate with conditions for one-half hour. Stop and restart the engine not less than five times. Operate all hydraulic controls for five-minute periods. Hold each hydraulic control in each actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the scraper through not less than 10 full right and 10 full left turns. Make not less than five abrupt panic stops from a speed not less than 20 mph using the service brake. Release and reapply the parking brake not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the scraper by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that scrapers previously accepted are similarly defective unless evidence satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.5.2.2 Sound level. The contractor shall perform sound level tests in accordance with SAE J88 and SAE J919. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.6.1 shall constitute failure of this demonstration.

4.5.2.3 Weight. The weight of the scrapers and scraper sections (type II only) shall be measured with a scale calibrated within 6 months of the demonstration. Inability to meet the maximum weight limits of 3.7.1 and 3.8 shall constitute failure of this demonstration

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4.5.2.4 ROPS/FOPS removal and replacement. The removal and replacement of the ROPS/FOPS canopy shall be demonstrated following procedures in the contractor's manual. Permanent deformation or damage to any part of the ROPS/FOPS canopy or its mounts, or inability to meet the requirements of 3.6.2 and 3.19 shall constitute failure of this demonstration.

4.5.2.5 Sectionalization and interchangeability. The type II scraper less ROPS/FOPS shall be demonstrated for compliance to the sectionalization requirements of 3.8 and the interchangeability requirements of 3.1. The demonstration shall consist of the following:

(a) Sectionalization. This demonstration shall be conducted on a relatively flat unimproved area of sufficient size for disassembly, maneuvering and reassembly of the scraper. Using hand tools furnished with the scraper (no power tools allowed), separate the scraper into two sections. Drive the powered section under its own power for a distance of 50 feet from the remaining section and steer the powered section at least 45 degrees to the longitudinal axis of the remaining section. Drive and steer the powered section to join with the remaining section and reassemble the two sections into a composite unit using the above hand tools only. Record the time required for the maneuvering and reassembly operation. No chains, cables, hoists, blocking, or other vehicles shall be used to assist in the disassembly, maneuvering, and reassembly operations, Nonconformance to 3.8 shall constitute failure of this demonstration. The sectionalization demonstration will be compared to the disassembly and assembly instructions specified in 3.21. In addition, those assemblies or components unique to the type II vehicle, or required as a result of the sectionalization feature, will be evaluated and verified as part of this test, to include disaaembly, examination, and reassembly. Maintenance functions, procedures, or tools peculiar to the type II vehicle, as well as publications associated with the specific feature of sectionalization, will also be evaluated and verified. All errors in publications or procedures shall be corrected by the contractor and submitted to the Government for approval prior to acceptance of preproduction vehicles. During this demonstration, the elevator raising and lowering functions and maintenance task shall be performed following procedures in the technical publications using soldiers serving as typical construction equipment operators and mechanics using common tools plus any special tools identified by the contractor.

(b) Interchangeability. This demonstration shall be conducted at any indoor or outdoor site selected by the contractor. Hoists, cables chains, and overhead crane may be used in this demonstration as required. eparate the tractor section from the scraper section. Connect the tractor section to the tanker section of the water distributor conforming to MIL-D-52987, connecting all air, hydraulic, and electric lines. Install the distributor air controls on the mounting provisions supplied with the tractor section. Operate the water distributor for at least 30 minutes, actuating all controls, including one complete fill and discharge of the tanker. Failure to complete the tractor/tanker connection, or any malfunction during operation shall constitute failure of this demonstration.

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4.5.2.6 Parachute suspension provisions. The parachute suspension provisions shall be demonstrated to prove conformance to 3.19.1. Apply a force of 1.5 times the limit load as specified in MIL-STD-814 to each of the parachute suspension provisions. Weld failure or permanent deformation of the scaper or suspension provisions shall constitute failure of this demonstration.

4.5.2.7 Sling provisions. The slinging provisons shall be demonstrated to prove conformance to 3.19.2. Nonconformance to 3.19.2, weld failure, or permanent deformation of the scraper or the slinging provisons shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

(a) Lift the scraper and bold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the scraper. Spreader bars are permitted. Determine the force and angles of application for each provison. Measure the clearance between each sling and the scraper.
(b) Restrain the scraper by anchoring the main frame and subject each slinging provision to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for aminimum of 90 seconds.
(c) Repeat (a) and (b) above for each section of the type II scraper.

4.5.2.8 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.19.3. Apply a force equal to the limit load as specified in MIL-STD-814 to each of the tiedown provisions. Weld failure or permanent deformation of the scraper or the tiedown provisions shall constitute failure of this demonstration.

4.5.2.9 Extraction prachute provison. The extraction parachute provision shall be demonstrated to prove conformance to 3.19.4. Apply a force equal to 1.5 times the limit load specified in 3.19.4 to t-he provision. Weld failure or permanent deformation of the scraper or the extraction parachute provison shall constitute failure of this demonstration.

4.5.2.10 Construction task. The scraper shall be demonstrated as follows for a total of 40 hours to prove conformance to 3.9 and 3.10.3: self-load, hauls dump and spread undisturbed or recompacted soil. The haul distances shall be at least 1500 feet. Record the time required for self-loading to heaped capacity at least 5 times. All lighting shall remain on throughout this demonstration. Nonconformance to 3.9 and 3.10.3 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this demonstration.

4.5.2.11 Airdrop. The scraper shall be subjected to simulated airdrop impact demonstration in accordance with MIL-STD-669 to prove conformance to 3.19. This demonstration will be conducted by the Government at a Government facility; however, acceptance of the preproduction models will be dependent on the successful completion of this demonstration. The demonstration shall consist of the following:

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(a) Rig the scraper in accordance with MIL-STD-669 and confirm that the rigged weight and dimensions conform to the requirements of AFSC Design Handbook DH1-11.
(b) Drop test the machine in accordance with MIL-STD-669.
(c) After the first drop, the machine shall be de-rigged, examined, started, and driven forward and backward a few feet, and all controls shall be actuated.
(d) The machine shall then be sectionalized and reassembled in accord- " ante with the contractor's instructions to ascertain that no damage was done to the sectionalization features by the test drop. (This step is omitted for the type I machine.)
(e) Re-rig and drop test the machine a second time in accordance with MIL-STD-669.
(f) After de-rigging, the scraper shall be demonstrated for 8 hours performing 20 percent of the construction task in 4.5.2.10.

Failure to meet the weight and dimensional requirements of AFSC Design Handbook DH1-11, weld failure, leakage, permanent deformation of any structural member or component, inability to reassemble the components removed for airdrop, inability to disassemble and reassemble the type II scraper, or inability of the scraper to perform its intended functions shall constitute failure of this demonstration.

4.5.2.12 Certification. The contractor shall certify in writing to the contracting officer that the scraper complies with the environmental condition requirements of 3.9.1, the stability requirements of 3.9.3, the fording requirements of 3.9.4, the engine requirements of 3.10, the horsepower requirements of 3.10.1 the steering requirements of 3.13.1, the braking system requirements of 3.13.2, and the rail shipment requirements of 3.19. If certification is not supplied, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations,

4.5.2.13 Certified test reports. The contractor shall furnish certified test reports to Prove compliance with the ROPS/FOPS requirements of 3.6.2, travel speed requirements of 3.9.2, electromagnetic radiation requirements of 3.16.4, and bowl capacity requirements of 3.18.1. If certified test reports are not available, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations.

4.6 Inspection of packaging The preservation, packing, and marking shall be examined to determine compliance with MIL-T-3351 and MIL-STD-129 (for level A or B) and MIL-STD-1188 (for commercial).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each complete scraper, components, and attachments shall be preserved in accordance with level A preservation and packaging requirments of MIL-T-3351.

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5.1.2 Level B. Each complete scraper, components, and attachments shall be preserved in accordance with the level B preservation and packaging requirements of MIL-T-3351.

5.1.3 Commercial. Each complete scraper and all items issued with the scraper shall be suitably preserved and packed in accordance with MIL-STD-1188 for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentization or carbonizing of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating in detail the preservation materials to be removed, in detail, and any other servicing required prior to placing the scraper in operation.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete scraper, preserved as specified in 5.1, shall be packed in accordance with level A packing requirements of MIL-T-3351, as specified for type II unboxed (mobile) scraper.

5.2.2 Level B. Each complete scraper, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-T-3351, as specified for type II unboxed (mobile) scraper.

5.2.3 Commercial. Each complete scraper preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188. Commercial packing must protect items against physical and environmental damage during shipment, handling and storage.

5.3 Marking.

5.3.1 Military. Marking for level A or B protection shall be in accordance with MIL-STD-129.

5.3.2 Commecial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The scraper covered by this specification is intended for use in general earthmoving work such as construction of military roads and airfields, and for airborne and airmobile operations.

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6.2 Ordering data. Procurement document should specify the following:

(a) Title, number and date of this specification.
(b) Type scrapers required (see 1.2).
(c) Time frame required for submission of the preproductin models (see 3.3).
(d) Technical publications required (see 3.26).
(e) Time frame for completing and submitting a Technical Information Package (see 3.29).
(f) Men Basic Issue Items are to be furnished (see 3.31).
(g) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.30.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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